PROGRAM:

```
//program for insertion sort
#include <stdio.h>
// Function to print the array elements
void printArray(int arr[], int size) {
   for (int i = 0; i < size; i++) {
      printf("%d ", arr[i]);
   }
   printf("\n");
}

// Function to implement insertion sort
void insertionSort(int arr[], int size) {
   int key, j;

   // Outer loop starts from the second element (index 1)
   for (int i = 1; i < size; i++) {
      key = arr[i]; // The element to be inserted into the sorted part
      j = i - 1;

      /* Move elements of arr[0..i-1] that are greater than the key
        to one position ahead of their current position */
      while (j >= 0 && arr[j] > key) {
         arr[j + 1] = arr[j]; // Shift element to the right
         j = j - 1;          // Move to the next element on the left
      }

      // Insert the key at its correct position
      arr[j + 1] = key;
      printArray(arr, size);

   }
}

int main() {
   int data[] = {12, 11, 13, 5, 6};
   int size = sizeof(data) / sizeof(data[0]);

   printf("Original array:\n");
   printArray(data, size);

   insertionSort(data, size);

   printf("Sorted array in ascending order:\n");
   printArray(data, size);
```

```
   return 0;
}
```

OUTPUT:

Original array:

12 11 13 5 6

11 12 13 5 6

11 12 13 5 6

5 11 12 13 6

5 6 11 12 13

Sorted array in ascending order:

5 6 11 12 13